

Carbon as a First-Class Signal: Designing a Standard for OpenTelemetry





Head of DevRel Engineering

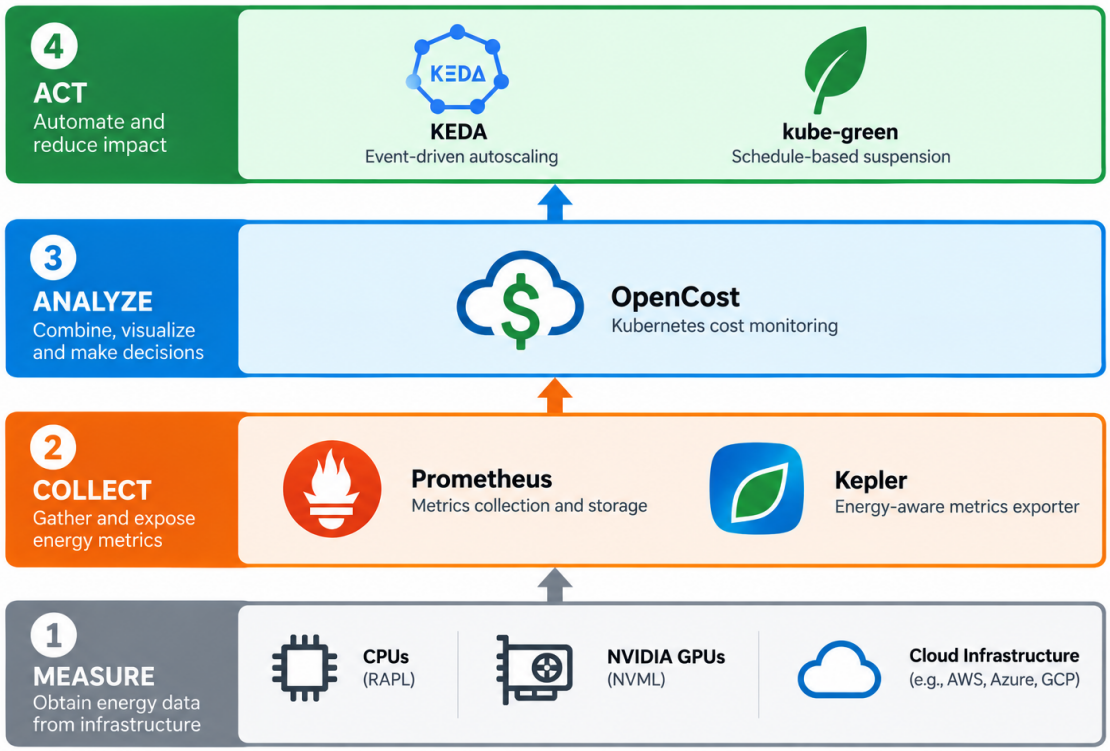
VictoriaMetrics

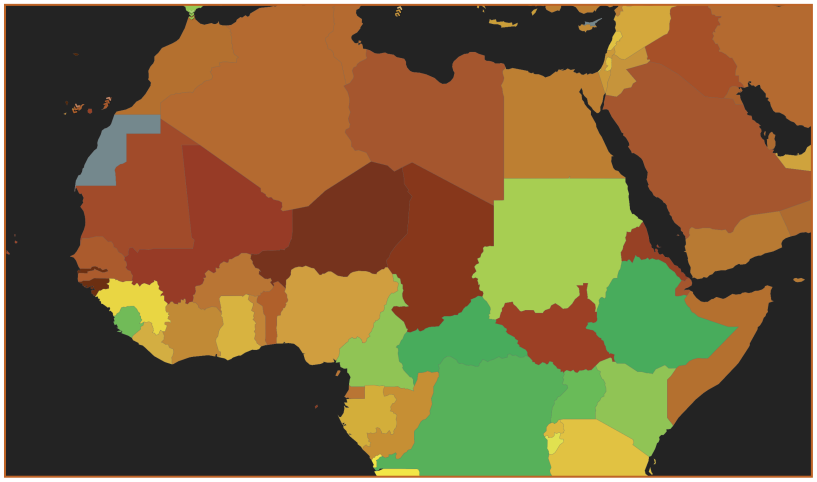
KEPLER

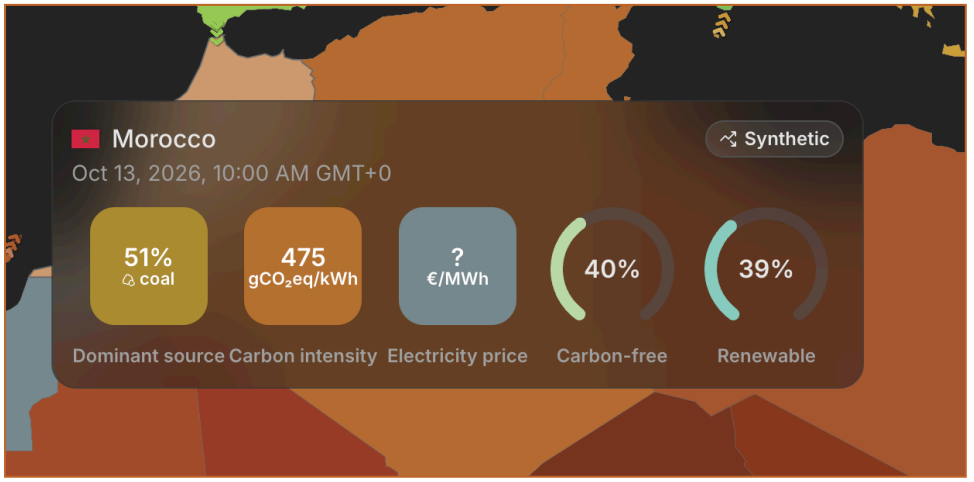
AMBASSADOR
CLOUD NATIVE
COMPUTING FOUNDATION

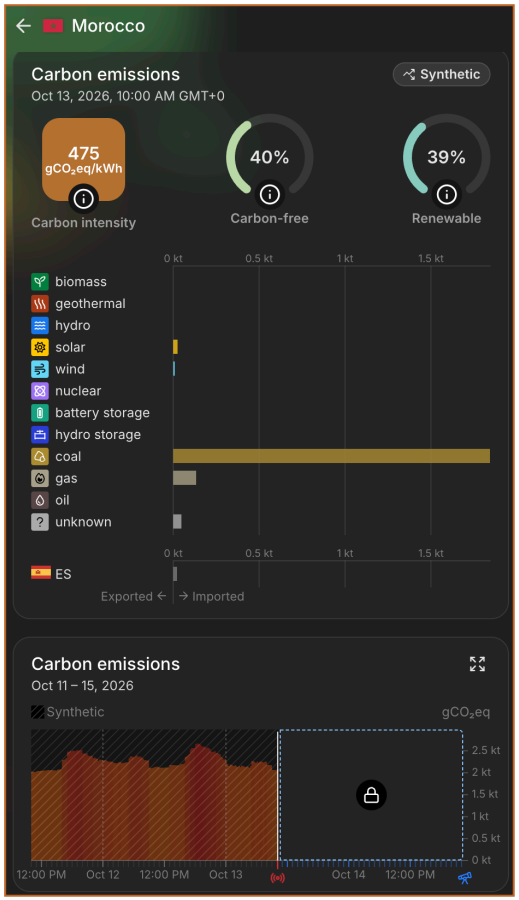
TAG
OPERATIONAL
RESILIENCE

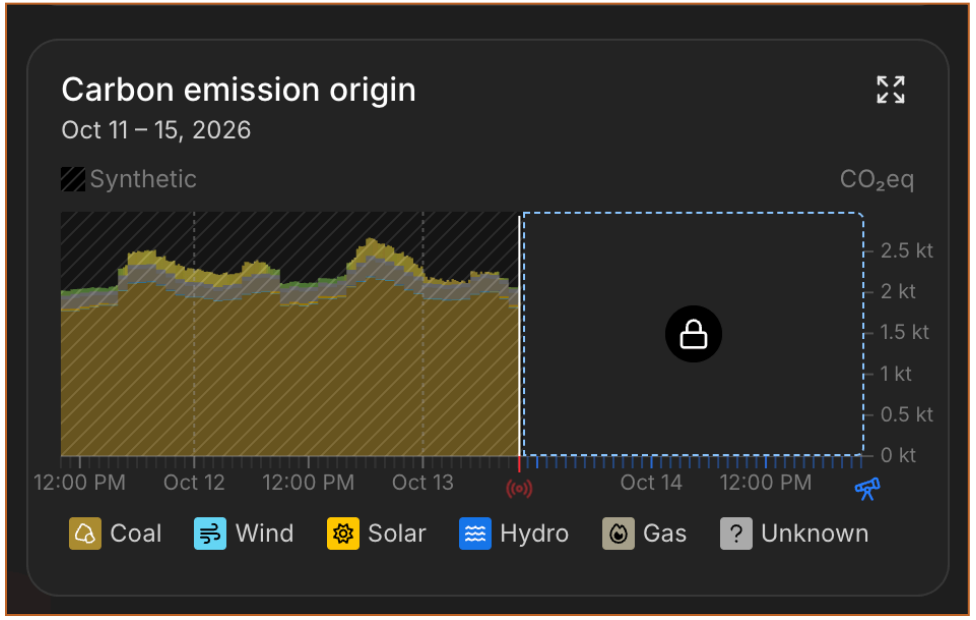
MERGE
FORWARD
CLOUD NATIVE
COMPUTING FOUNDATION


4
ACT
Automate and reduce impact
KEDA
KEDA
Event-driven autoscaling
kube-green
Schedule-based suspension
3
ANALYZE
Combine, visualize and make decisions
OpenCost
Kubernetes cost monitoring
2
COLLECT
Gather and expose energy metrics
Prometheus
Metrics collection and storage
Kepler
Energy-aware metrics exporter
1
MEASURE
Obtain energy data from infrastructure
CPUs
(RAPL)
NVIDIA GPUs
(NVML)
Cloud Infrastructure
(e.g., AWS, Azure, GCP)









Electricity Maps: https://app.electricitymaps.com/

The missing signal

Software Carbon Intensity (SCI)
= carbon emissions per functional unit

Version A	Version B
100 kg CO_2e 1 million transactions	**110 kg CO_2e** 2 million transactions
0.0001 kg CO_2e per transaction	**0.000055 kg CO_2e** per transaction

Software Carbon Intensity (SCI)

Green Software Foundation

$$SCI = ((E \times I) + M) \text{ per } R$$

E
Energy

Energy consumed by the software

I
Intensity

Carbon intensity of the electricity grid

M
Embodied

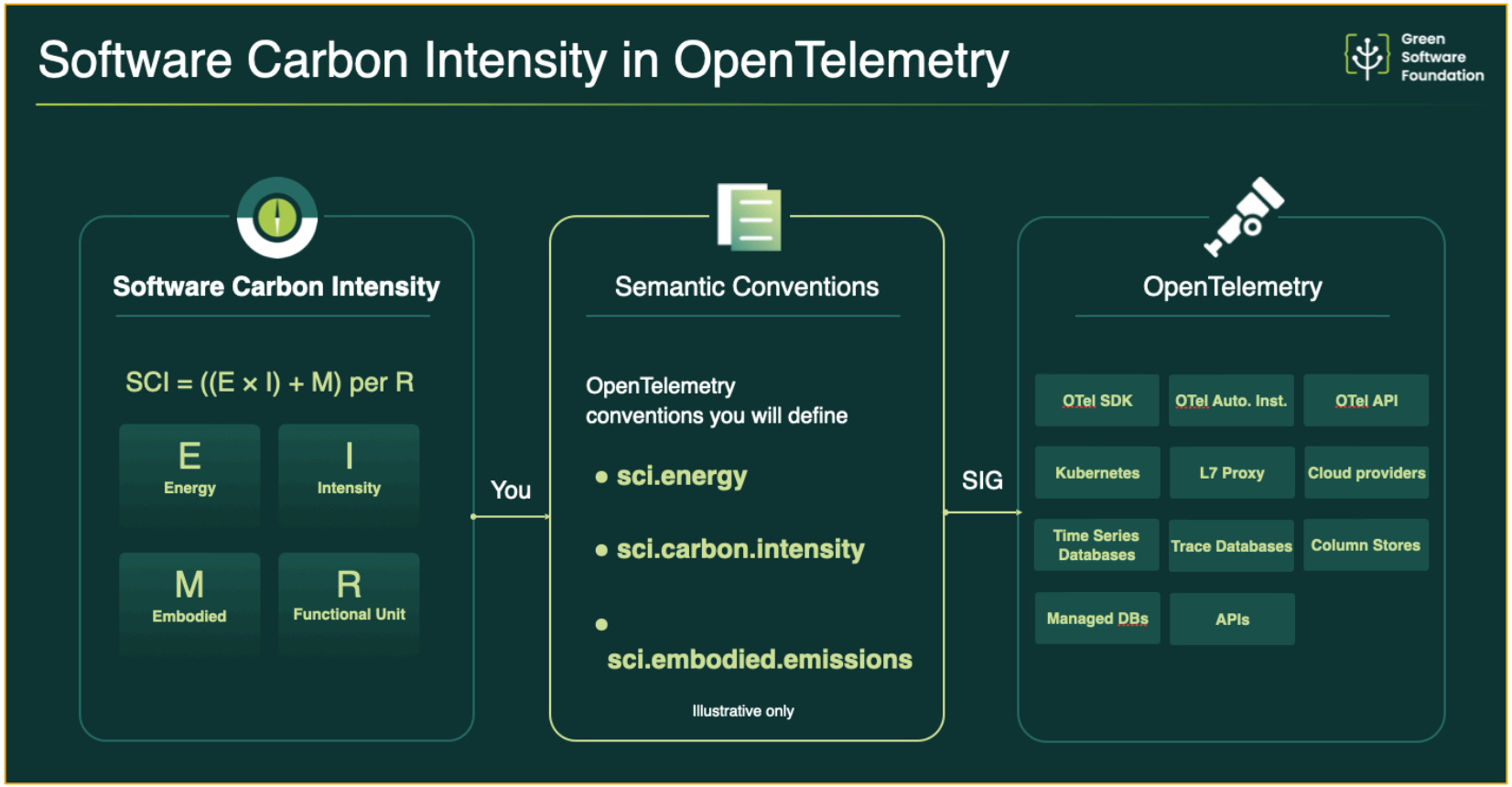
Emissions baked into the hardware

R
Functional Unit

Rate — per transaction, user, request

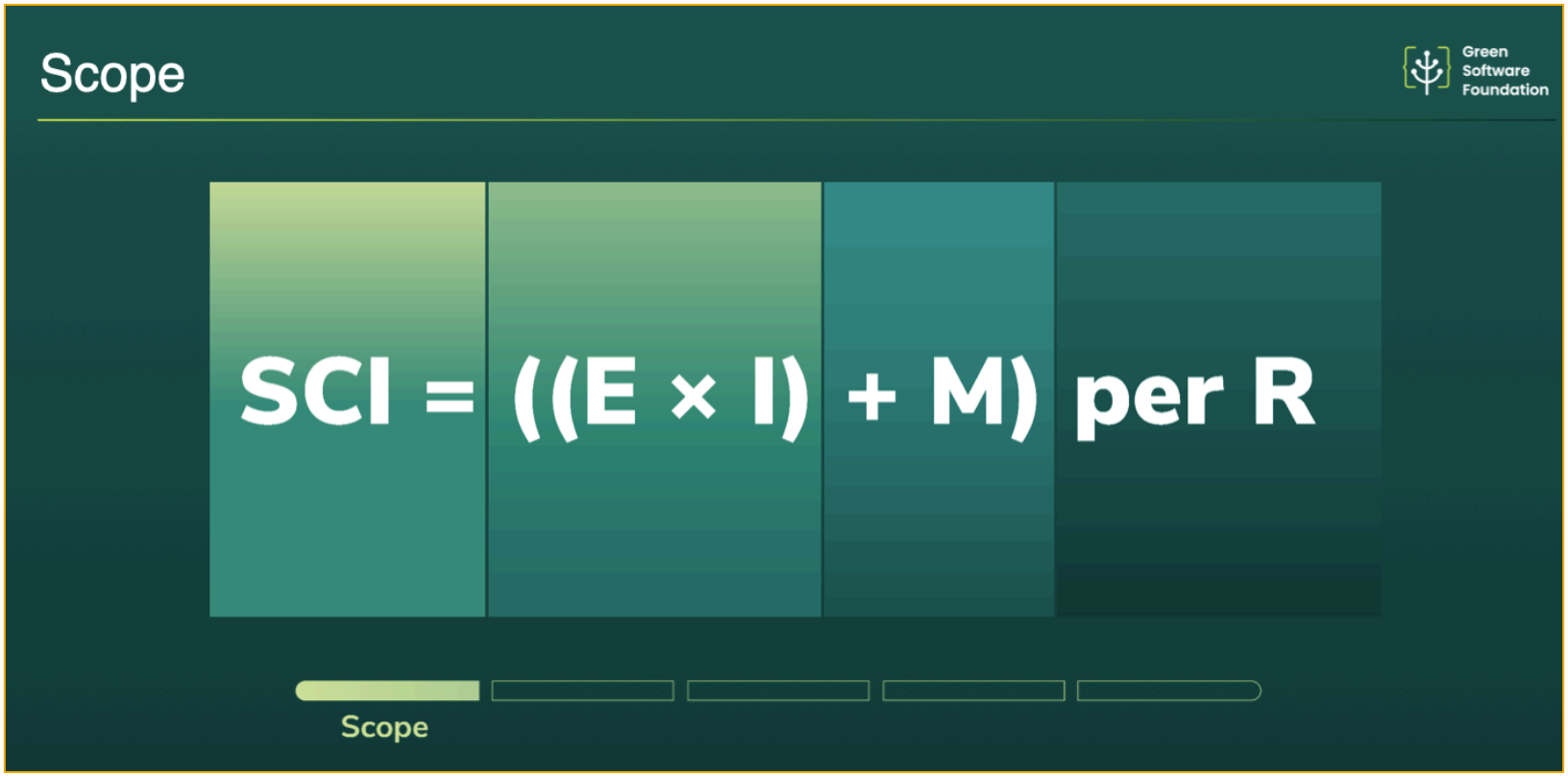
Operational Energy

ISO/IEC 21031:2024 I Designed for Action not Reporting

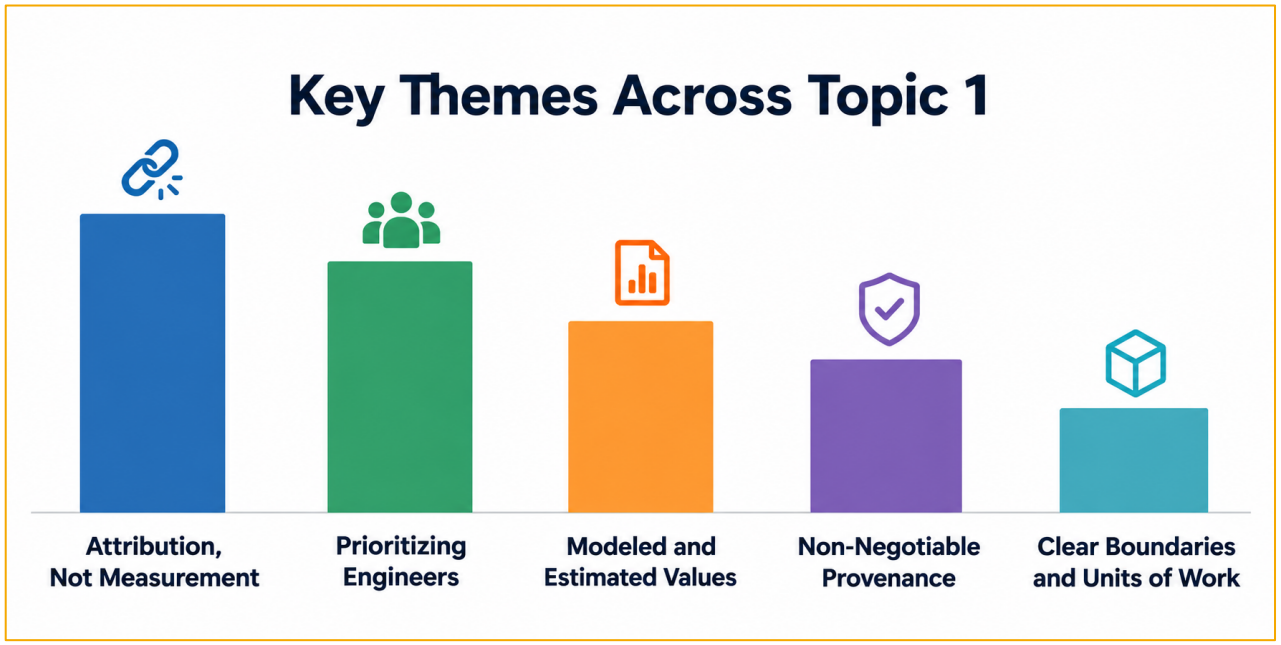


Source: https://github.com/Green-Software-Foundation/sci-otel

Current work



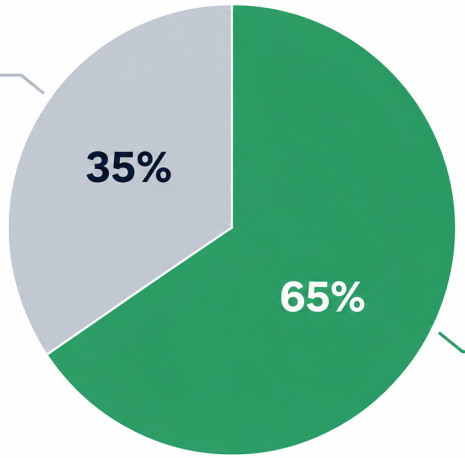
SCI for OpenTelemetry Assembly Results: Topic 1 - Scope & Use Cases



Source: https://github.com/Green-Software-Foundation/sci-otel/blob/assembly-topic-1/assembly/topics/1/topic-1-consensus-candidate.md

Participant Engagement

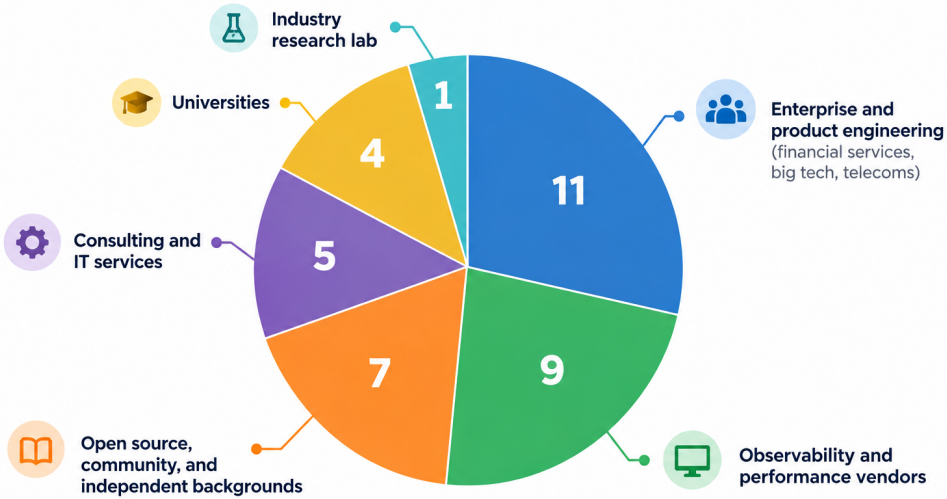
37 participants invited



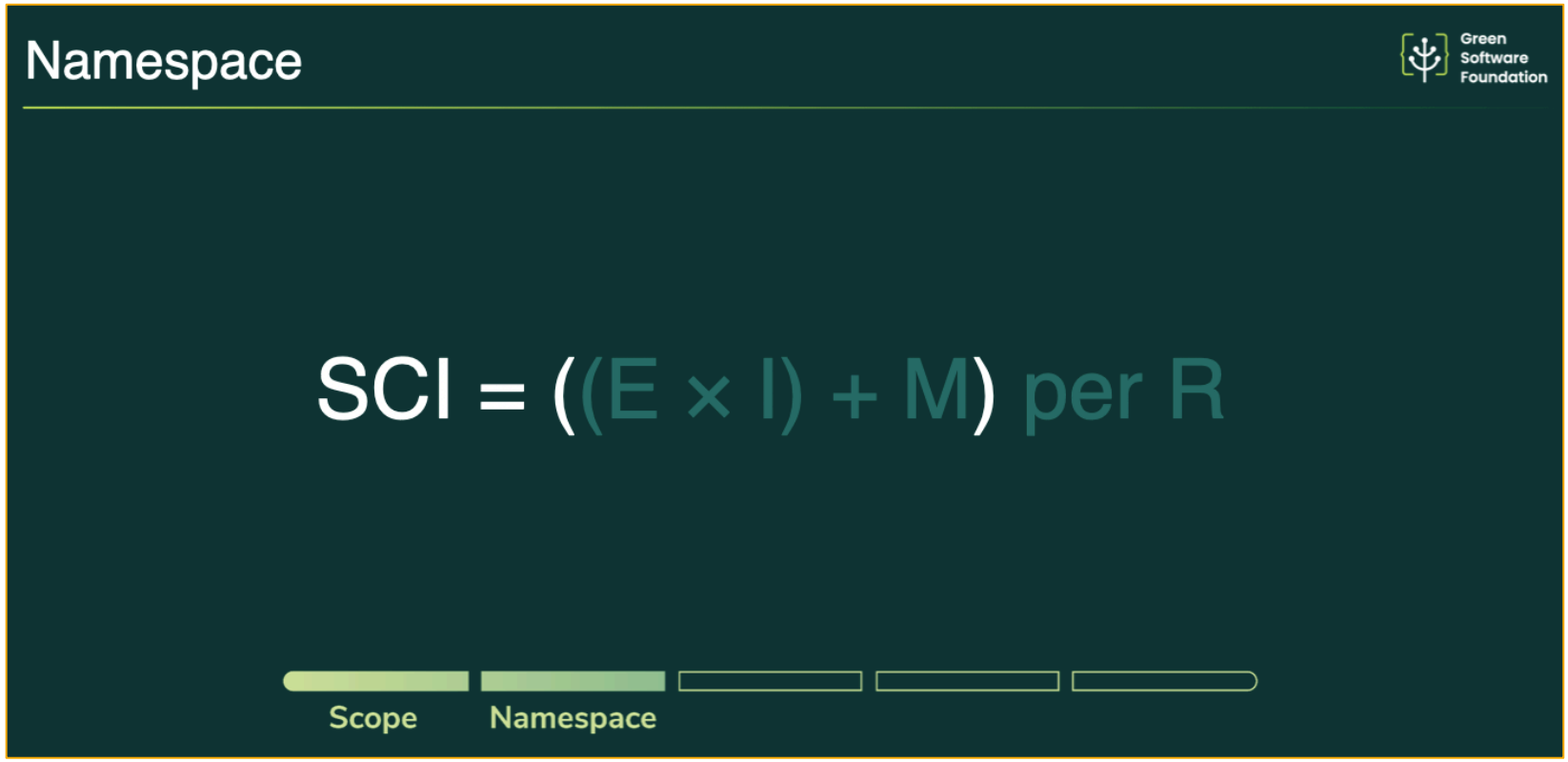
13
Did not participate

24
Participated across Discovery, Deliberate & Decide

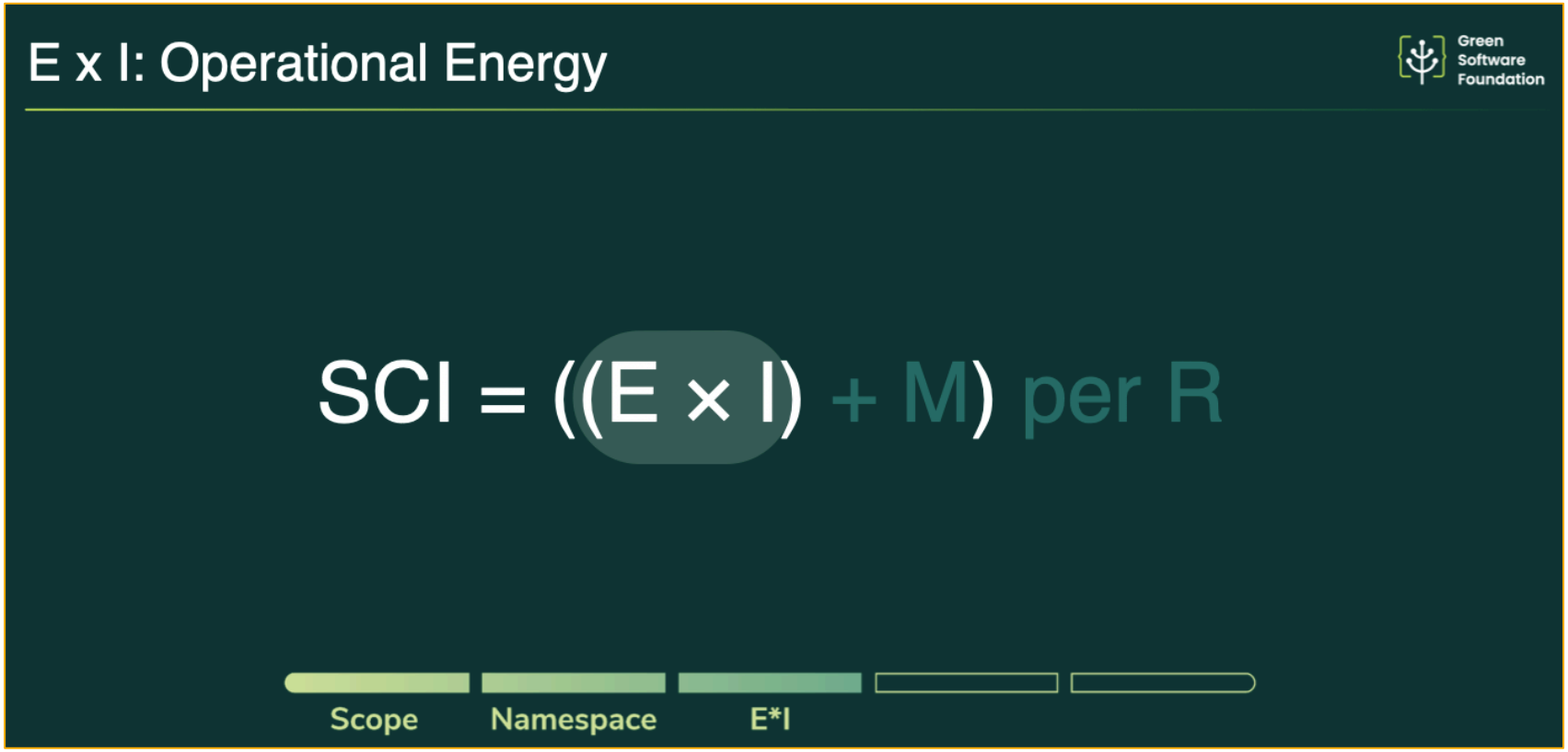
Participant Diversity



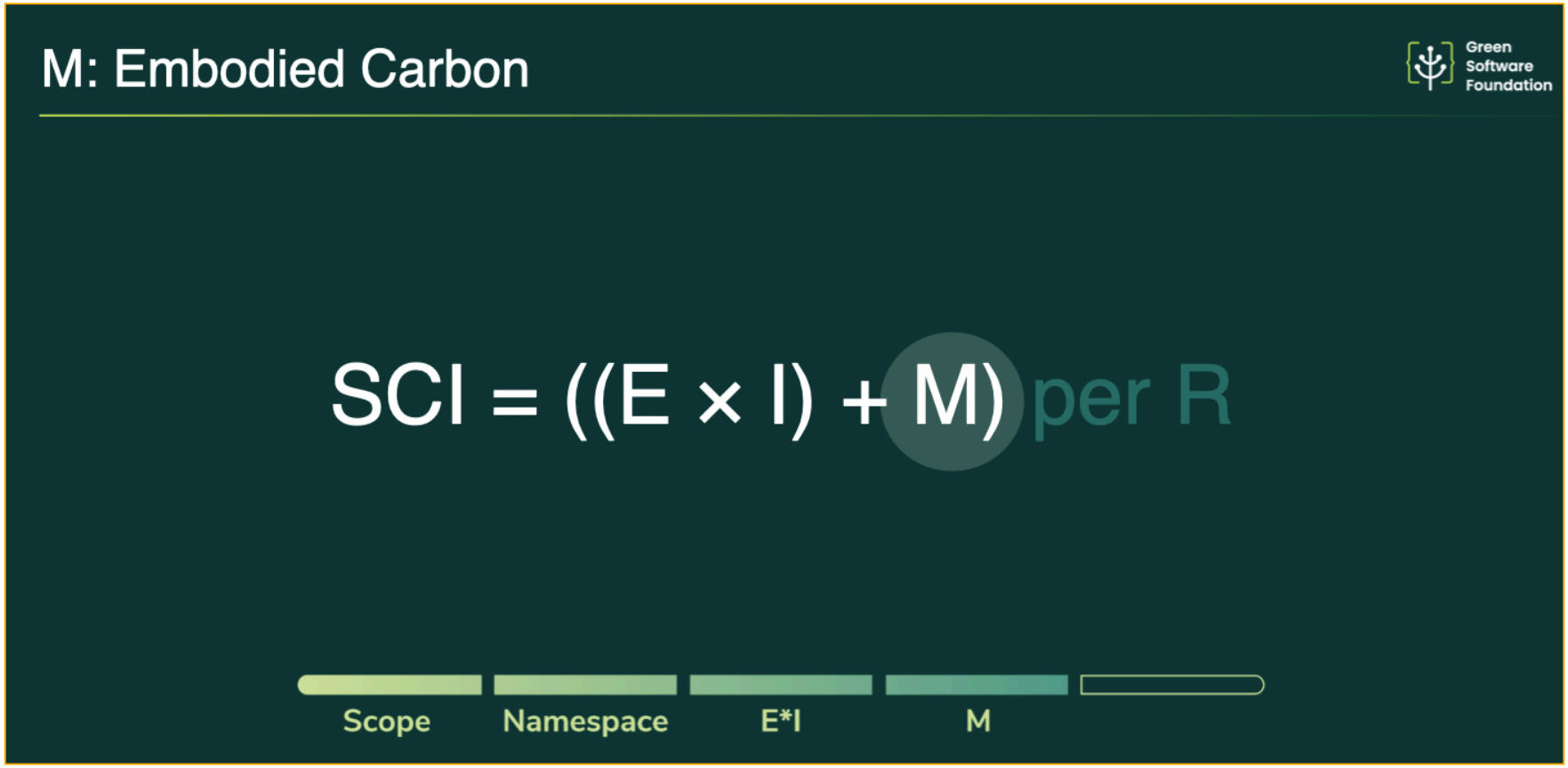
Coming up next!



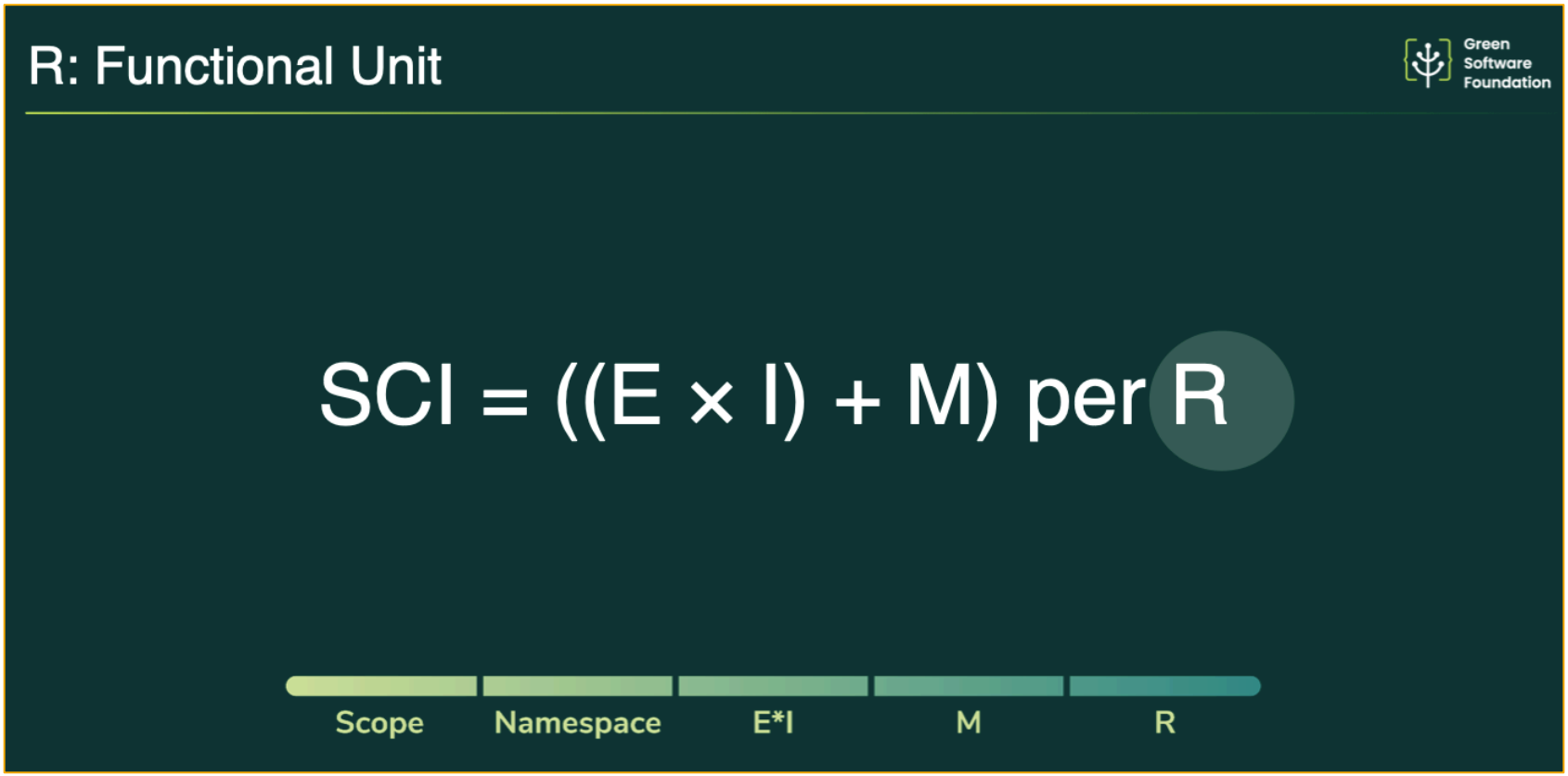
Coming up next!



Coming up next!

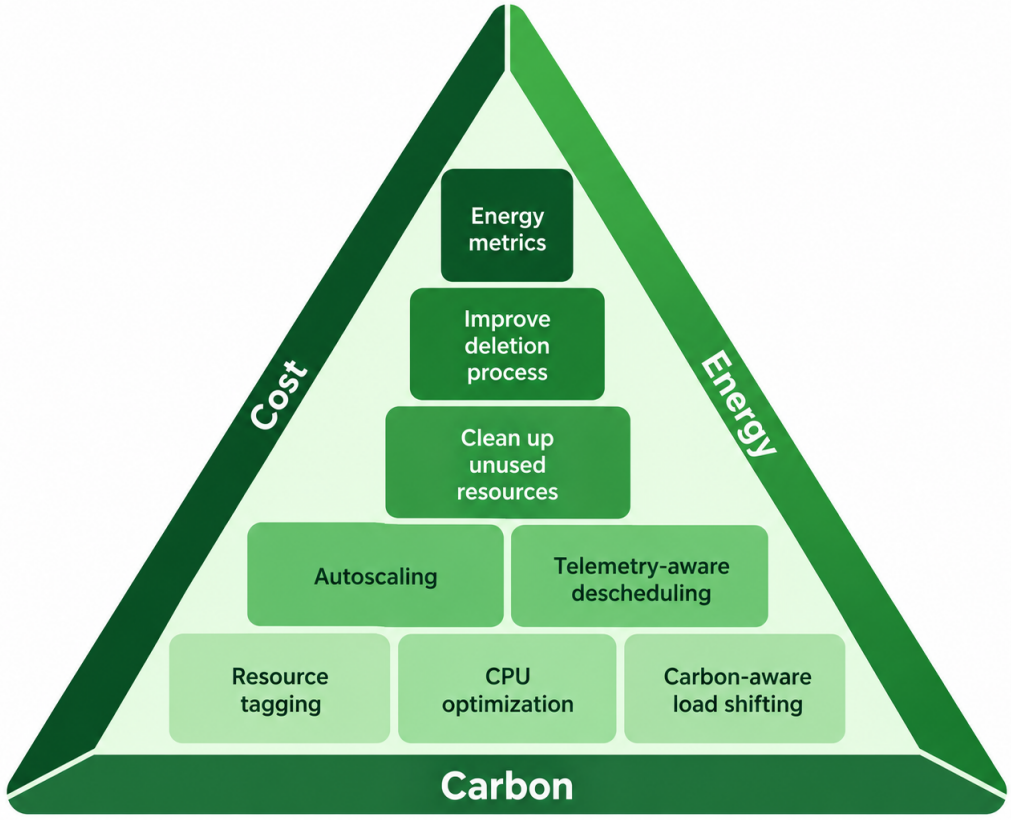


Coming up next!



What if carbon became context for the engineering decisions we already make?

GitOps + FinOps = GreenOps



Carbon is becoming another dimension of system efficiency

1. Measure it

Use the systems you already have to measure energy and carbon information.

2. Connect it to cost and performance

Link carbon and energy data to cost, performance and workloads.

3. Use it in engineering decisions

Ask whether this context changes an engineering decision and make carbon part of the conversation.



SCI gives us a way to measure.
OpenTelemetry gives us a common language.
Carbon can become a first-class operational signal.

Source: https://greensoftware.foundation/assemblies/sci-for-open-telemetry/



CNCF TAG Operational Resilience

Making cloud native systems more reliable, observable and efficient beyond deployment.

WHAT WE FOCUS ON

Observability • Reliability • Day 2 operations
Performance • Cost & resource efficiency
Energy & sustainability • Business continuity

HOW TO CONTRIBUTE

Join open community meetings
Contribute to initiatives and guides
Share use cases, reviews and lessons learned

JOIN US!

#tag-operational-resilience

slack.cncf.io · contribute.cncf.io/community/tags/operational-resilience/

Resources

https://www.linkedin.com/company/cncf-environmental-sustainability-tcg/
https://ocgroups.dev/cncf/group/env
https://ocgroups.dev/cncf/group/q8h223s/event/4wqs2ud
https://sylvaproject.org/
https://github.com/sustainable-computing-io/kepler/issues

Thank you!



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